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# EMERGING TECHNOLOGIES:
# Adoption of Emerging Technologies and Its Impact on Modern Organization

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# I. Introduction

Emerging technologies are newly developing innovations that continue to evolve, playing a crucial role in modern organizations. People define these technologies in various ways; for instance, some view them as the evolution or refinement of existing systems. Artificial Intelligence is a prime example: although first developed in the 1950s, it remains an emerging technology because it is constantly advancing and finding new applications across diverse fields. These advancements have the potential to reshape how businesses, governments, and individuals operate. By enhancing decision-making, improving efficiency, and driving cost savings, emerging technologies unlock new opportunities and streamline workflows.

Consequently, organizations must understand these innovations to evaluate its practical use, this report will discuss the 4 major technological development: Big Data Analytics, Internet of Things (IoT), Cloud Computing, and Augmented Reality.

# II. Four Major Technological Development

### Big Data Analytics

Big Data Analytics is the process of collecting, organizing, and studying large amounts of data from different sources to find patterns and useful information. It helps businesses make quick and better decisions, especially in production and customer management. With modern technology, large amounts of data can be processed quickly, allowing organizations to get useful answers and information in real time.

Big Data Analytics application:

- Tracking Customer Spending Habit, Shopping Behavior
- Recommendation
- Smart Traffic System
- Secure Air Traffic System
- Auto Driving Car
- Virtual Personal Assistant Tool
- Online Educational Courses
- Media and Entertainment

**Example:** An online shopping platform can use AI to analyze a customer's previous purchases and recommend products that may interest them.

## Internet of Things (IoT)

The Internet of Things (IoT) refers to everyday physical devices that are connected to the Internet and can communicate with other devices. These devices can use sensors and wireless connections to collect and share information. IoT allows devices to work together and send information to a central system. This helps organizations receive information quickly and make decisions or take action in real time.

Internet of Things (IoT) application:

- Smart Agriculture
- Smart Vehicles
- Smart Home
- Smart Healthcare
- Smart Cities

**Example:** When someone receives a message on their phone, their smartwatch can show a notification without needing to check the phone directly.

## Cloud Computing

Cloud computing refers to providing services such as storage, applications, and other computer resources through the Internet. As cloud technology continues to improve, it can provide faster responses and allow data to be processed almost instantly. More devices and systems are also using the cloud to store and manage data, which helps businesses provide better and more data-based services.

Cloud Computing application:

- Online Data Storage
- Backup and Recovery
- BigData Analysis
- Testing and Developement
- E-commerces Application
- E-governance Application
- Medical fields

**Example:** A patient visits a hospital, the doctor can access their previous records, test results, and other important information through a cloud-based system.

### Augmented Reality

Augmented Reality (AR) can help workers by showing useful information through their phones, tablets, or AR devices. For example, workers in a warehouse can use AR to quickly find and select the correct parts. It can also provide repair instructions while they are working. In the future, more companies may use AR to give workers real-time information, helping them make better decisions and complete their tasks more easily.

Augmented reality application:

- Manufacturing
- Medical field
- Use of AR Glasses
- Classroom Education

**Example:** IKEA AR app allows users to use their phone's camera to see how furniture may look in their actual room. Users can choose a piece of furniture and place a digital version of it on the screen while viewing their real surroundings.

## III. COMPARISON

| EMERGING TECHNOLOGIES | MAIN PURPOSE | APPLICATION | EXAMPLE |
|---|---|---|---|
| Big data Analytics | Analyzes large amount of data to find useful information. | Costumer behavior, traffic systems, and recommendation. | Online shopping recommendations. |
| Internet of Things | Connects physical devices to the Internet. | Smart homes, healthcare, agriculture. | Smartwatch connected to a smartphone |
| Cloud Computing | Provides storage and computer services through the Internet. | Data storage, backup, e-commerce, healthcare. | Cloud-based hospital records. |
| Augmented Reality | Adds digital objects or information to the real world. | Education, manufacturing, shopping. | IKEA AR app. |

# IV. RECOMMENDATION

Based on the four emerging technologies discussed, organizations should consider using technologies that can help improve their daily operations and services. Before adopting a technology, they should first identify their needs and make sure that the technology is suitable for their organization.

Big Data Analytics can be used to understand customer behavior and help organizations make better decisions. IoT can be useful for connecting devices and collecting information in real time. Cloud Computing can help organizations store and access their data more easily, while Augmented Reality can help workers and customers by providing useful information through digital devices.

Organizations should also consider the cost, security, privacy, and skills needed when using these technologies. Employees should be given proper training so they can use the technology correctly. It is also better to test the technology first before using it on a larger scale.

# V. CONCLUSION

Emerging technologies are becoming an important part of modern organizations. The four technologies discussed in this report—Big Data Analytics, Internet of Things, Cloud Computing, and Augmented Reality—can help organizations improve their work and provide better services.

Big Data Analytics helps organizations understand and use large amounts of information. IoT allows different devices to connect and communicate with each other. Cloud Computing makes it easier to store and access data through the Internet, while Augmented Reality provides digital information that can be used together with the real world.

In my own understanding, these technologies can make work easier, faster, and more efficient when they are used properly. However, organizations should not adopt new technology just because it is popular. They should first consider their needs, resources, security, and the possible effects of using it. As technology continues to develop, learning about these emerging technologies can help organizations prepare for the future.

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